

03rd Sep, 2025

C.W

Wed.

Chp 2: Computational

Thinking

E₁

Algorithms.

Q1: How many problem we discussed in computing?

Ans: There are three problems:

1) Decision Problem:

- A decision problem occurs when a given input requires a binary response.
- Responses may take True / False.
- e.g:-
 - The problem of even / odd.
 - The problem of prime number.

2) Search Problem:

- In science & engineering, many problems are solved using search.
- e.g:-
 - finding a path between two cities.

3) Counting Problem:

- These work on the principle that if an event / decision has a number of choices A & another has B number of choices. Then the number of possible unique combinations would be $A \times B$.

Example:

No. of Shirts: 5

No. of Pants: 3

Total possible casual dresses:

= No. of shirts $\times$ no. of pants.

= 5×3

= 15

Example 2:

Monitors = 4

Keyboards = 2

CPU / Computer = 4

Printer = 3

Total possible computers systems:

$$= 4 \times 2 \times 4 \times 3$$

$$= 96.$$

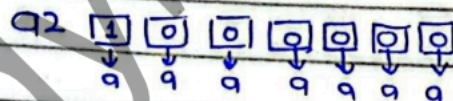
Q1: The telephone number - - - - -
- - - - - given area code.

Telephone number $\rightarrow$ 9 digits

1st two digits are fixed for area code

Next 7 numbers

It cannot begin / start with 0.

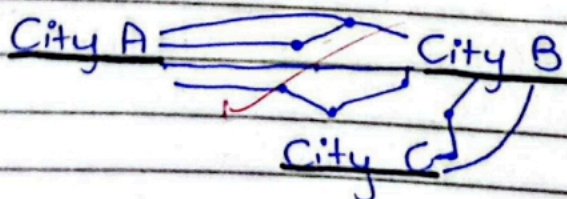


Total possible combinations: $1 \times 1 \times 9 \times 10 \times 10 \times 10 \times 10 \times 10 \times 10$

$$= 9 \times 10^6 \text{ Ans}$$

$$(9,000,000)$$

Q3: From city A to B - - - - -
- - - - - by city B?



How many routes are going from City A to City C = ?

Total routes = $4 \times 2 = 8$ Ans.

Q24 A student has
..... he has to take?

Physics = $(P_1, P_2, P_3) = 3$

Science = $(S_1, S_2) = 2$

Mathematics = $(M_1, M_2) = 2$

Total ways to select a pair of course = $3 \times 2 \times 2$
= 12 Ans.

05th Sep, 2025

M.H.W

Chp 2: Computational Thinking

&

Algorithms

Q: What is the difference between solving simple problems and complex problems.

Ans:

Simple Problems Sets

i) Define and analyze a problem.

2) Design a plan.

3) Implement the plan.

4) Evaluate.

Complex Problems Sets.

i) Define and analyze a problem.

2) Decompose the problem.

3) Identify potential plans for each sub problem.

4) Select & design best plan or evaluate.

Unit: 2
Computational Thinking
E
Algorithms

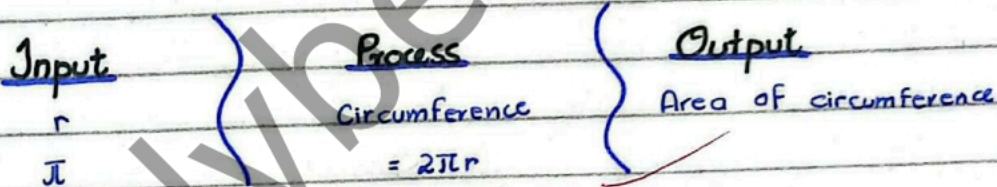
Q: Describe the steps of OPI Model.

Ans: The analyzing of problems allows us to break down the problem into three steps.

- 1) Input: input that need to be given to the system
- 2) Process: operations the system to perform.
- 3) Output: output to be presented.

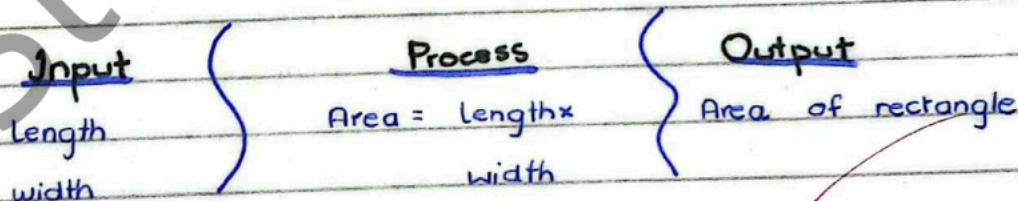
Q: Find the circumference. Draw IPO chart for this problem.

Ans: Area of circumference = $2\pi r$

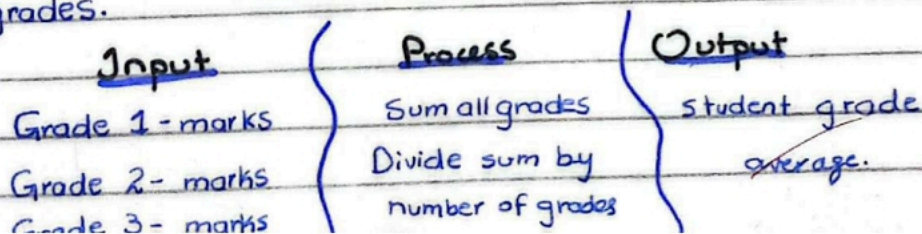


Example 2: IPO chart to compute area of rectangle.

Area = Length x width



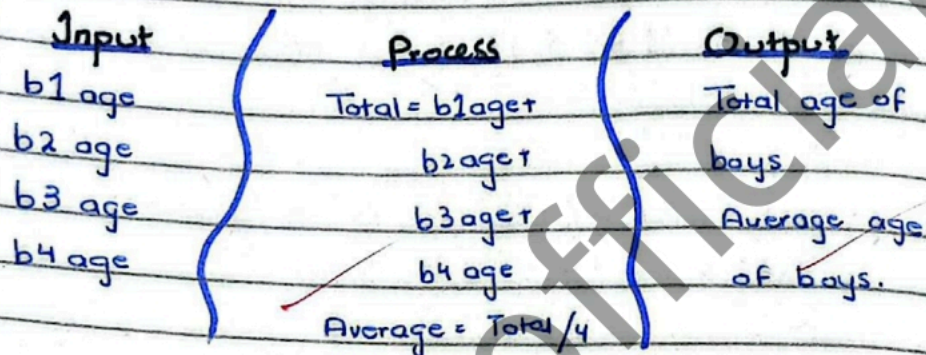
Example 3: Calculate a students average based on it's his grades.



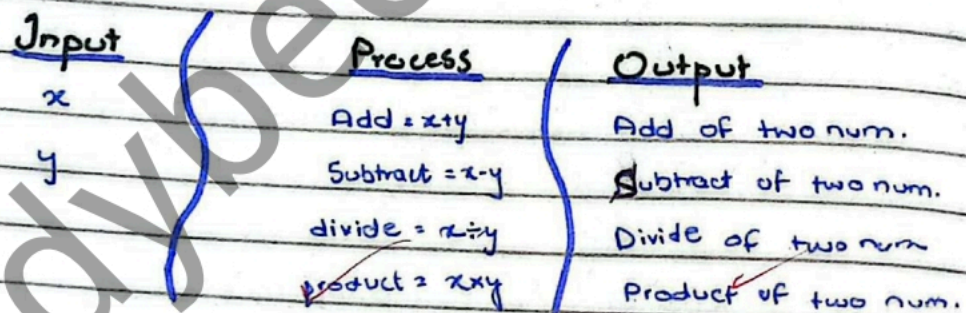
$$\text{Sum} = \text{Grade 1-marks} + \text{Grade 2-marks} + \text{Grade 3-marks}$$

$$\text{Average} = \text{Sum}/3$$

Q: Take 4 boys ages. Find total age? Find average age?
Output $\rightarrow$ Total & Average age.



Q: Input two no.s as x, y
 $\rightarrow$ Process $\rightarrow$ Add, subtract, division, Product



Q2: Find the exponents
given number.

Ans: a) Find the exponents of given number.
Calculate a^b using a power function.

b) Print odd numbers from 1 to 100 loop from 1 to 100
loop, print if odd.

c) Print the sequence in descending order

Sort 27, 24, 21, 18, 15, 12, 9, 6, 3, 0 descending and print.

d) Find the sum of even numbers up to 100.
Sum = 2550.

e) Print a multiplication table of a given number.

Loop from 1 to 10, print number * Loop counter.

10-09-2025

C.W

Weld

Unit 2: Computational Thinking

E₁

Algorithms

Q: What is computational thinking? Discuss the properties of computational thinking?

Q4: Short Questions: Write four properties of computational thinking?

Ans: Computational thinking:

→ It is problem solving method that computer scientists use as a skill to solve complex problems in a logical and systematic manner.

Properties:

① Decomposition:

→ Breaking down the larger problems into smaller ones & working on them one by one.

② Abstraction:

→ Remove unnecessary details from the problem.

③ Pattern Recognition:

→ Examine the problem to find previous solutions.

④ Algorithm design:

→ This involves creating step-by-step plan of the problem.

Q2: Why software designers prefer to use IPO charts?

Ans: Software designers used IPO charts because it break down the complex problem into three parts. Input, Process and output of problem classified in design phase.

Q2: Differentiate b/w computational thinking and logical thinking? (1mp)

Computational Thinking

1- Breaking down complex problems into steps that a computer can understand and solve.

2- Focuses on algorithmic problem solving for computers.

Logical Thinking

1- Using reasoning and logic to evaluate information, make decisions, and solve problems.

2- Logical thinking is a broader cognitive skill applicable to various problem-solving contexts.

10/09/25

checked